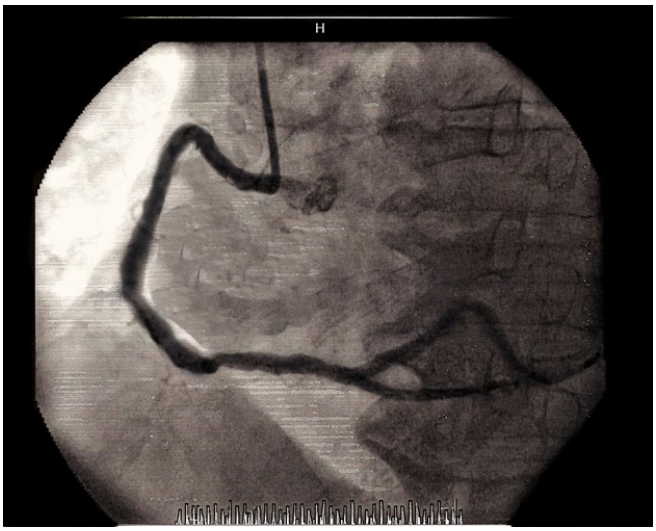
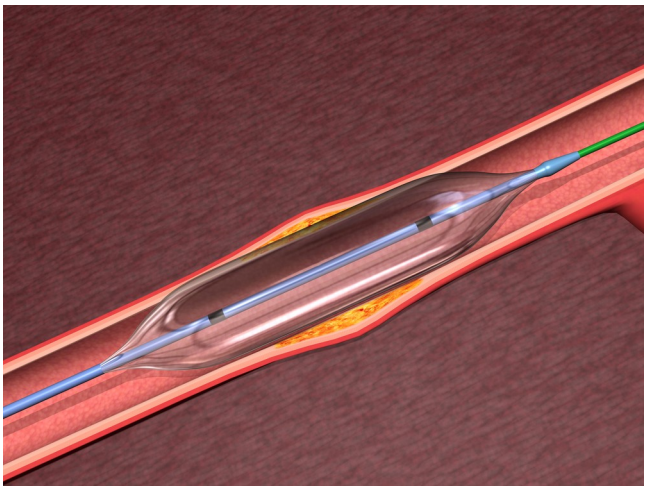
Cardiac catheterization was originally developed for performing a diagnostic arteriogram, or visualization of the arteries within the heart, but has been expanded in recent decades to include stenting of arteries, electrophysiologic studies, ablations, and placement of pacemakers and internal defibrillators.

Typically, patients will tolerate these procedures with light sedation, but as the procedures become more prolonged and complex, general anesthesia is becoming more common. Cardiology labs are often a long distance from the operating rooms and anesthesia supply center, which means the anesthetist must ensure that all reasonable equipment and supplies are close at hand.



A cardiac catheterization is performed by passing a catheter up the arterial tree (usually via a femoral or brachial artery) and into the aorta. Contrast media is then injection into the coronary arteries. Standard anesthesia monitors are used and for awake procedures, oxygen is administered by nasal cannula. The blood pressure can be monitored by non-invasive means as well as by direct monitoring via the arterial introducer. Heparin is frequently administered for a target ACT greater than 300 seconds and reversed at the end of the procedure with protamine.

A coronary stenosis greater than 50-70% is considered significant and may require percutaneous transluminal coronary angioplasty to increase flow through the artery. The patient may require analgesia during the inflation of the balloon, which can produce angina by occluding the artery completely. Other techniques used to increase blood flow through the artery include the placement of metallic stents to maintain arterial patency and atherectomy using a shaver or an excimer laser.



The risk of complications from cardiac catheterization is less than 2%. Major risks include cardiac ischemia, vessel trauma, coronary dissection, thromboembolic events, and renal damage or allergic reaction related to the use of contrast dye. Diabetics are particularly susceptible to renal damage from the overuse of radio-opaque contrast dye.

Miller RD. Miller's Anesthesia. 7th ed. Philadelphia, PA: Churchill-Livingston Elsevier; 2010: 2470-2472.

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